Matteo Palmonari

matteo.palmonari@disco.unimib.it

Beyond Query Answering

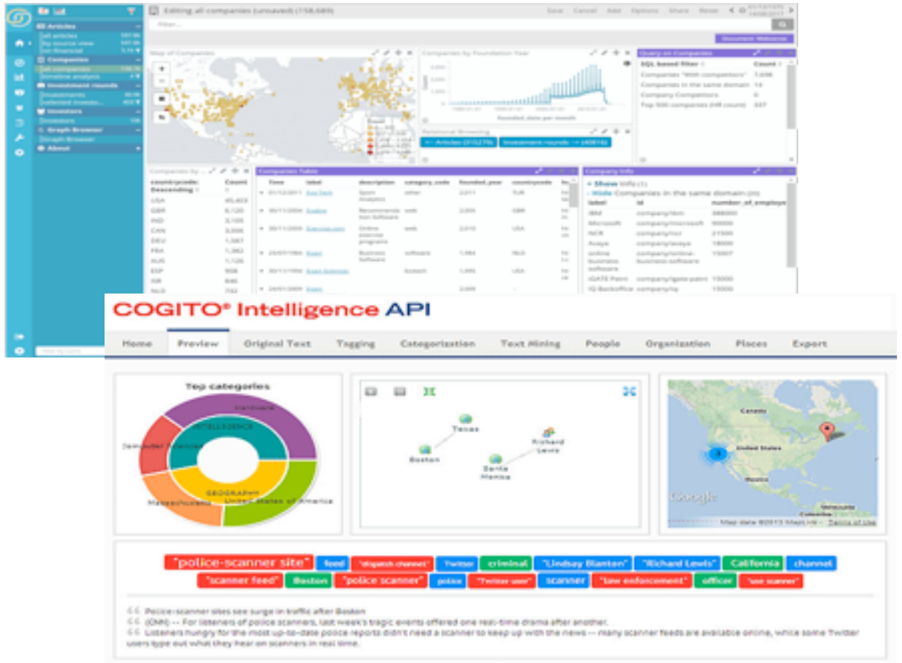
from Exploratory Search to (Content-based) Recommender Systems

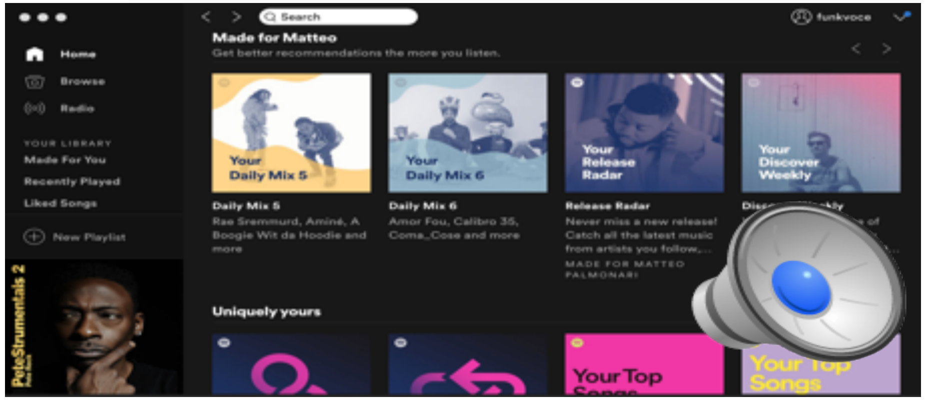
ITIS Lab – Innovative Technologies for Interaction and Services

Dipartimento di Informatica, Sistemistica e Comunicazione
Università degli Studi di Milano-Bicocca

Beyond Query Answering

- Structured queries represent specific information needs
  - You need to know what you are searching for to formulate the query
- Access to information beyond querying
  - **Passive** applications: support exploratory search, i.e., let users browse through the data before formulating a precise query
    - Discussed through a short presentation on supporting exploration of large Knowledge Bases
  - **Active** applications: push content to the user, i.e., predict what the user is interested in
    - Discussed by introducing a paradigmatic case: recommended systems

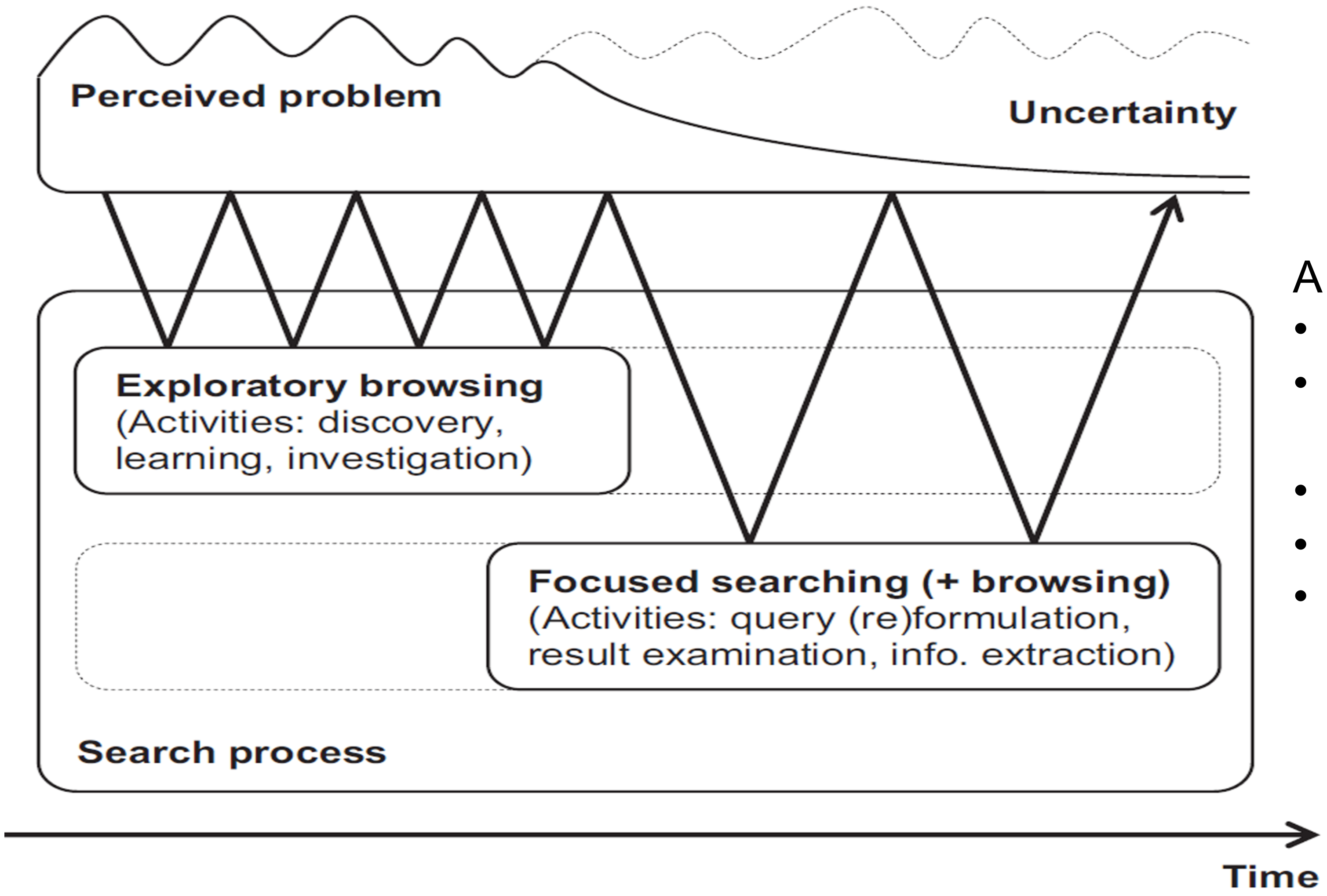




3rd important aspect of semantics: (inference + similarity +) relevance

Exploratory Search in and with Large Knowledge Bases

Beyond Query: Exploratory Search (White & Roth)



A system for exploratory search should:

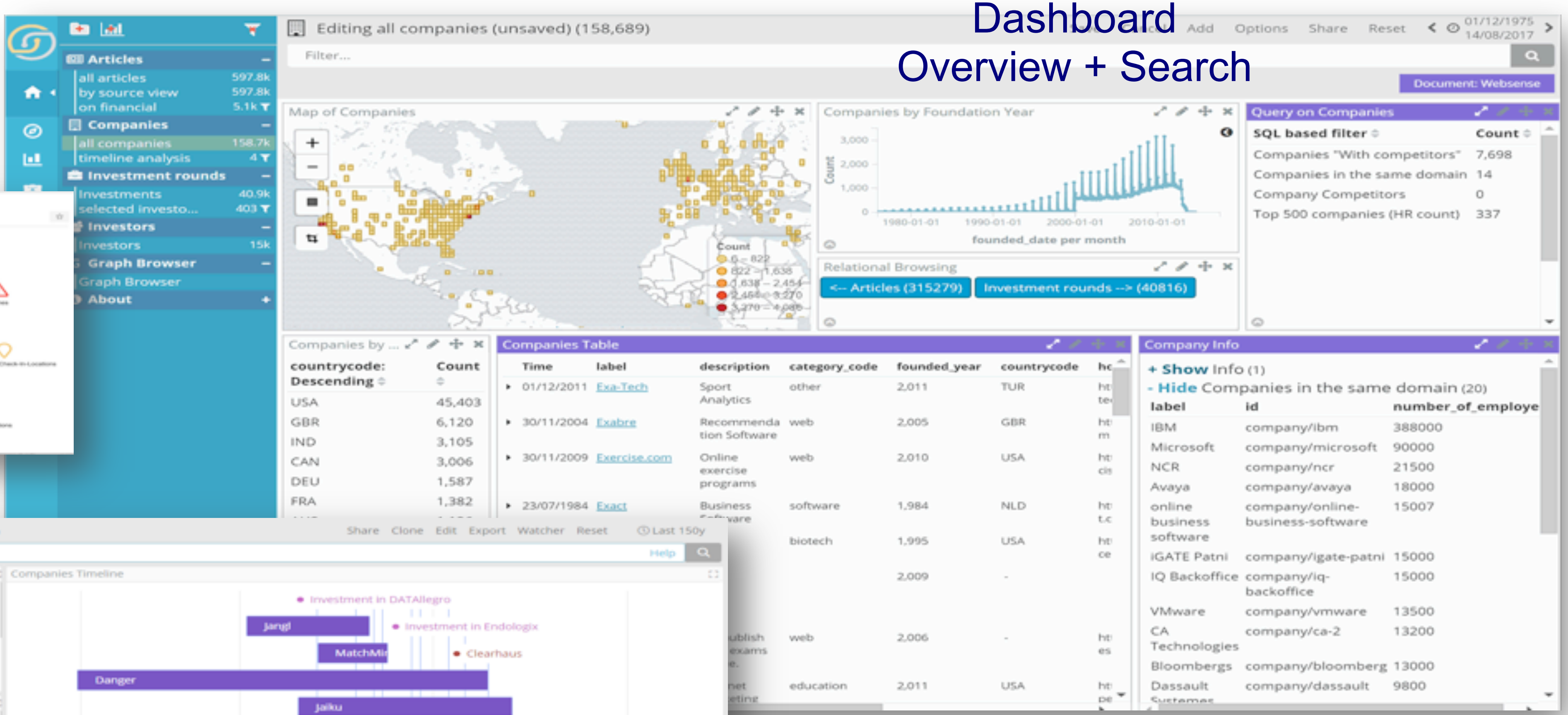
- Support query refining and filtering
- Disambiguate and rank results using context
- Use visualizations to facilitate insights
- Support collaboration
- Consider user history

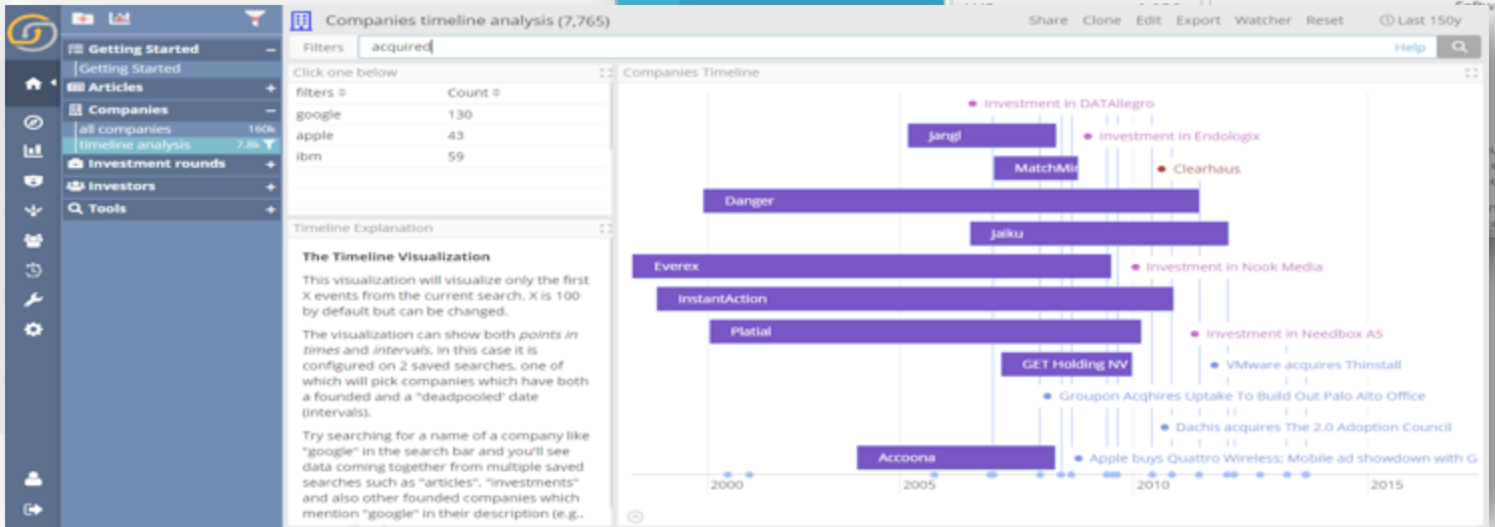
Ryen W White and Resa A Roth. Exploratory search: beyond the query-response paradigm (synthesis lectures on information concepts, retrieval & services). Morgan and Claypool Publishers, 3:9–23, 2009.

SIREN: a Modern Investigative Intelligence Platform

Graph

Dashboard
Overview + Search



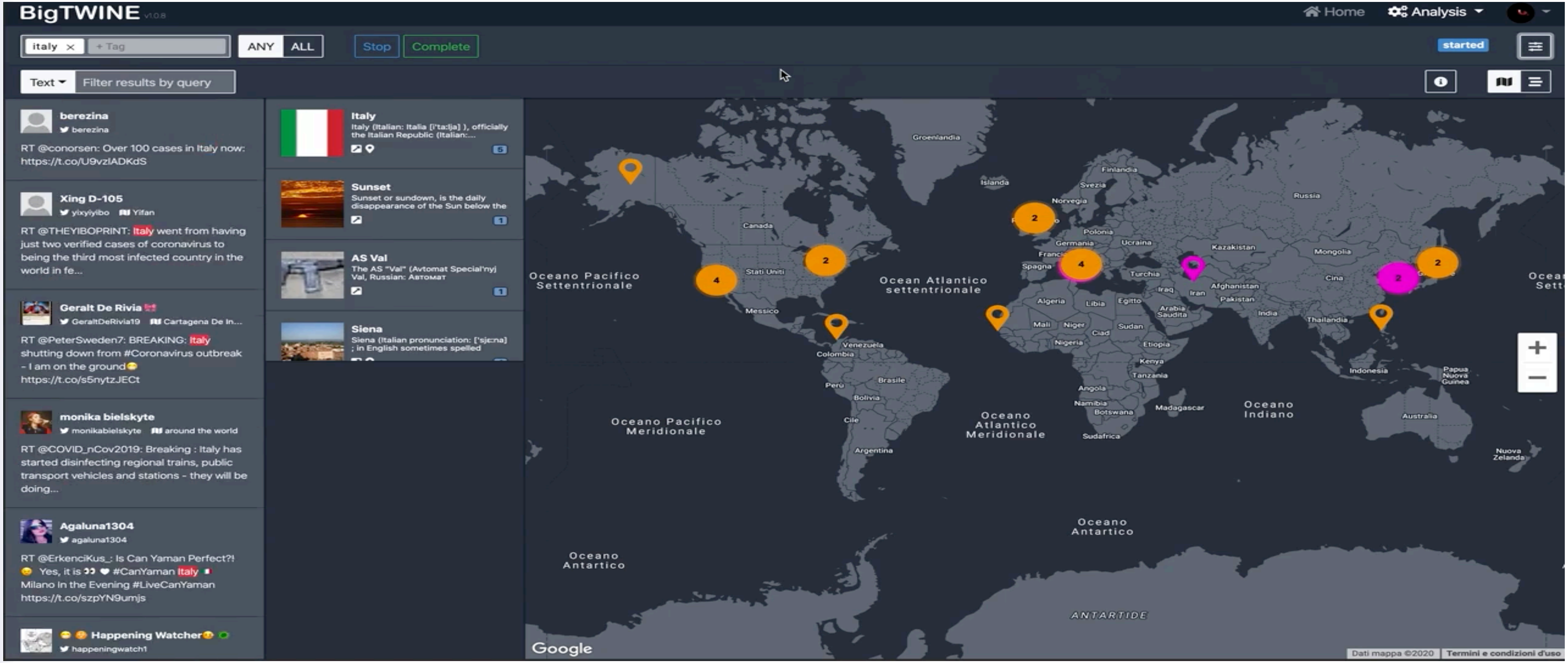


Timelines

Check more info online at:
https://siren.io/platform-overview/

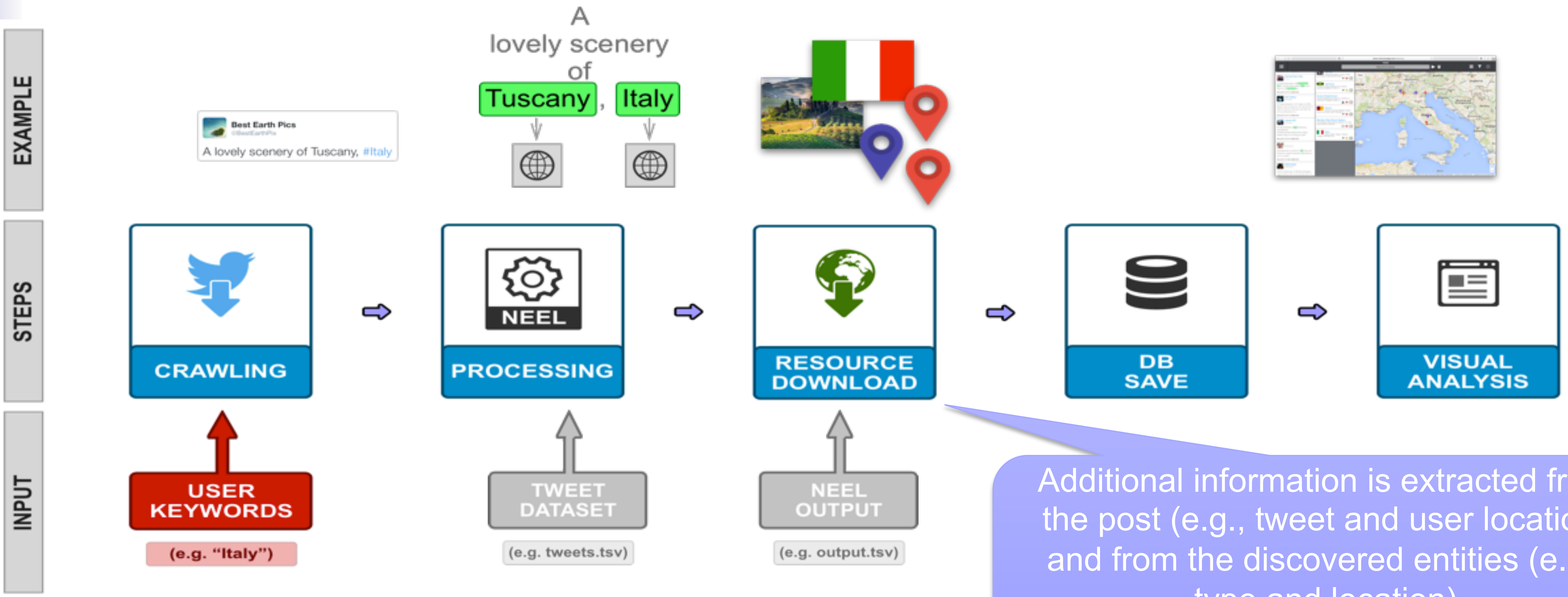
Expert System's COGITO

BigTWINE – BIG DATA PLATFORM FOR SEMANTIC ANALYSIS OF SOCIAL MEDIA



Check video at: https://www.youtube.com/watch?v=xCJBunDtxMs&feature=youtu.be

CONCEPTUAL PIPELINE – (Small) TWINE 1



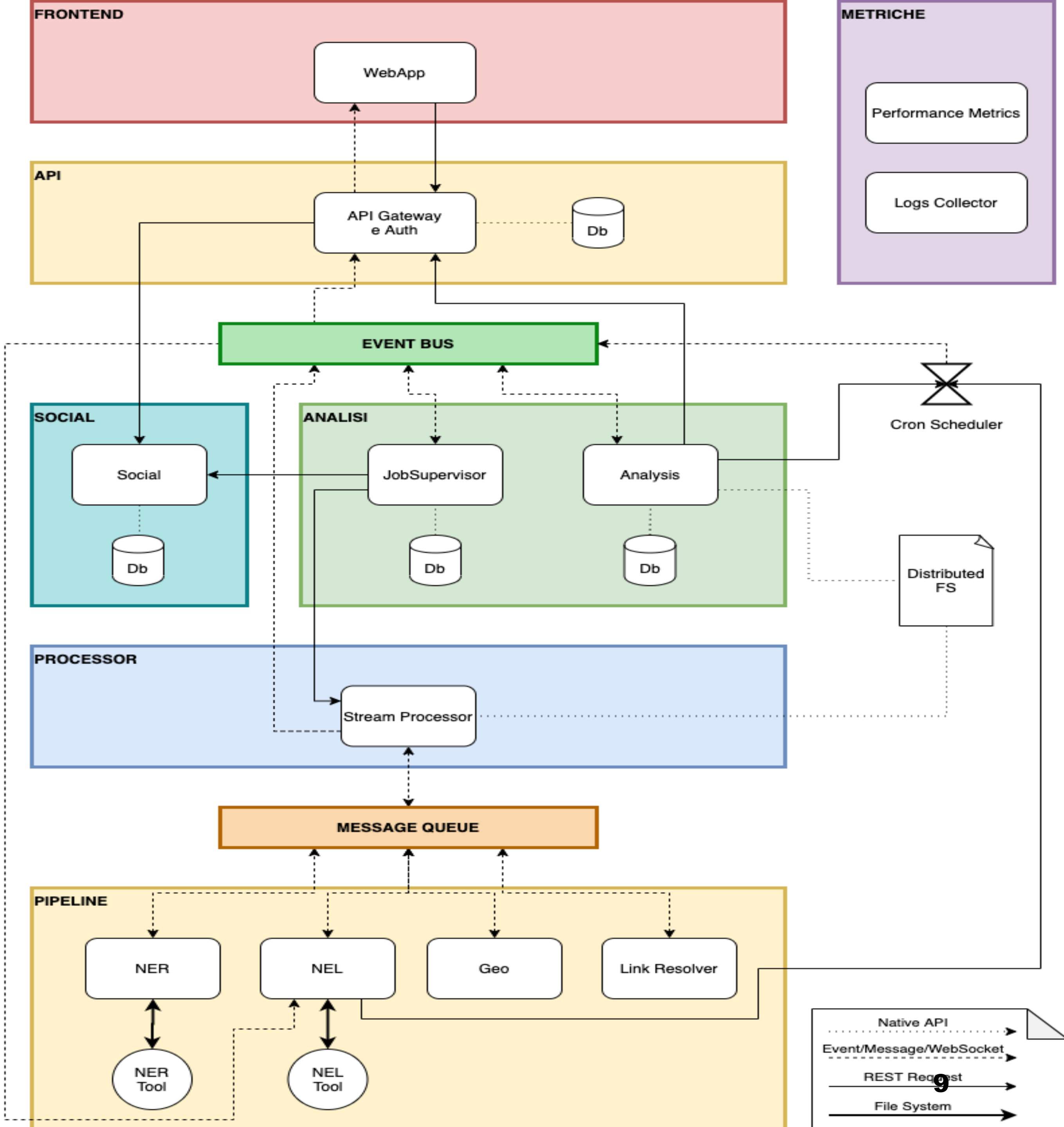
Additional information is extracted from the post (e.g., tweet and user location) and from the discovered entities (e.g., type and location)

This info supports the analysis

BIG TWINE is the result of the collaboration between the **MIND Lab** (E. Fersini, D. Nozza, V. Messina) and the **INSID&S Lab** (M. Palmonari, M. Ciavotta)

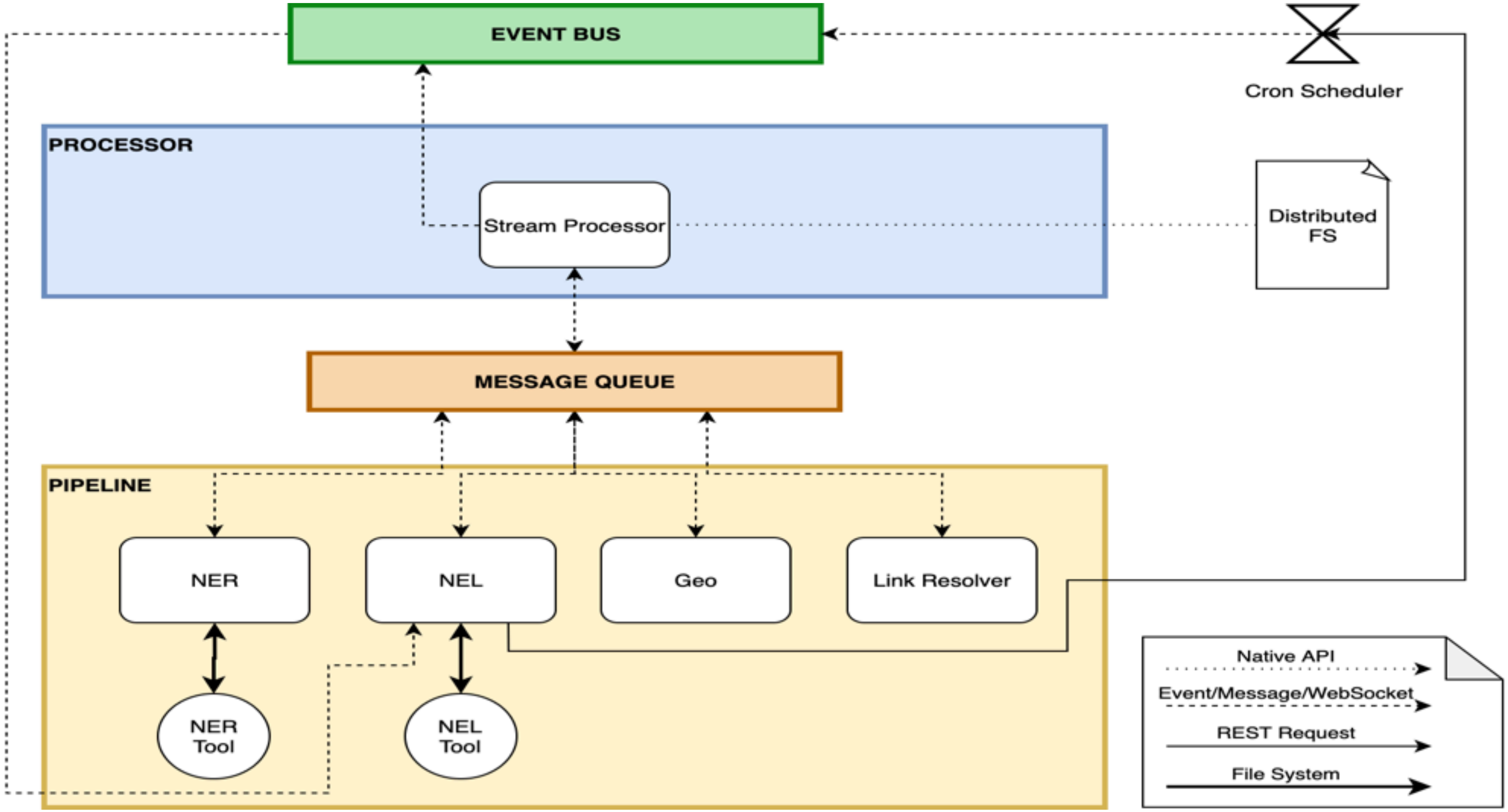
BigTWINE ARCHITECTURE

- Collection of services that communicate mostly in asynchronous way
- Mix of different technologies



Adapted from Fausto Ristagno's MA Thesis Presentation

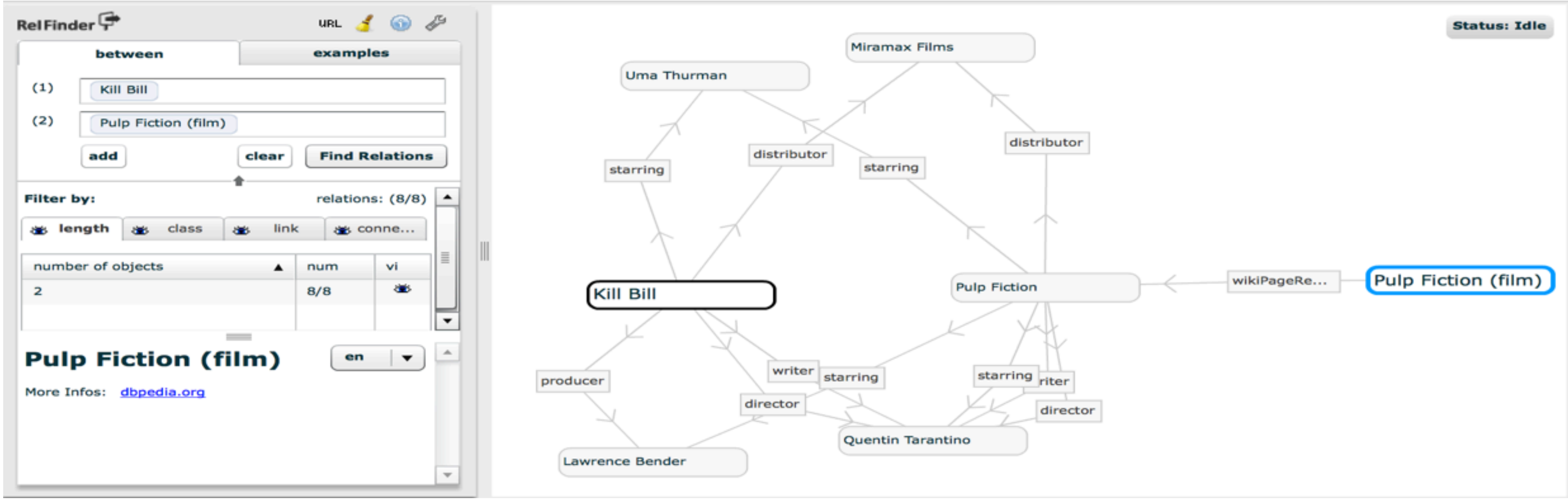
BigTWINE PIPELINE



Adapted from Fausto Ristagno's MA Thesis Presentation.
For more info about the architecture contact michele.ciavotta@unimib.it

Explorative Search of Relations in KGs

RelFinder



Philipp Heim, Steffen Lohmann and Timo Stegemann. Interactive Relationship Discovery via the Semantic Web. In: Proceedings of ESWC 2010, volume 6088, pages 303-317. Springer, 2010.

Exploration of Facts Relevant to a Text of Interest via Visual Interface

DaCena

Interactive exploration via visual interface

A Threat to Spanish Democracy

By CAYETANA ÁLVAREZ DE TOLEDO, NÚRIA AMAT and MARIO VARGAS LLOSA

NOV. 7, 2014

Original Article

MAIN ENTITY: **CATALONIA**

On Sunday, in defiance of an order from the Spanish Constitutional Court, a farcical referendum on Catalan **separatism** will take place. It is not clear who is holding it, as there are neither official voters' rolls nor auditors, only a semiofficial cabal of volunteers.All this in the name of "democracy" and the right to decide, but without respect for the rule of law or the true will of the people, and without acknowledging the gravity of the consequences that the secession of **Catalonia** would occasion.This supposed referendum — now rebranded a

Serendipity ?

relevant — unexpected

Threshold ?

300 / 40107

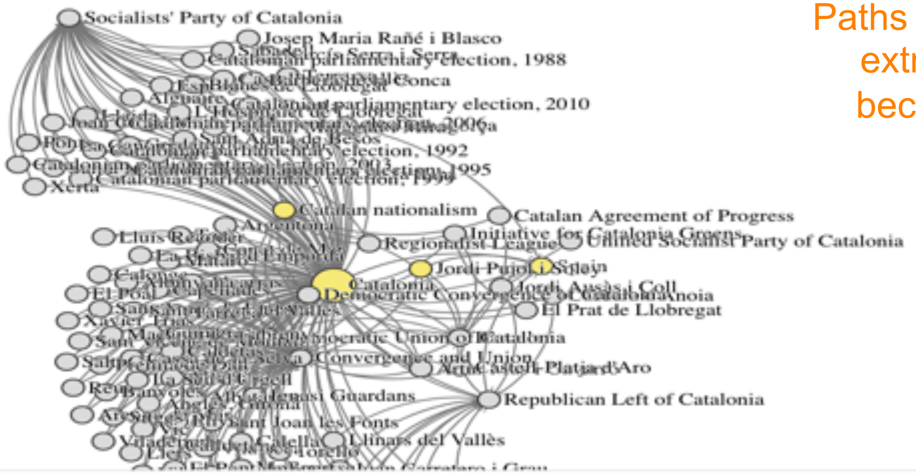
Path length

1 2 3

SHOW THE COMPLETE NETWORK

Overview

Details

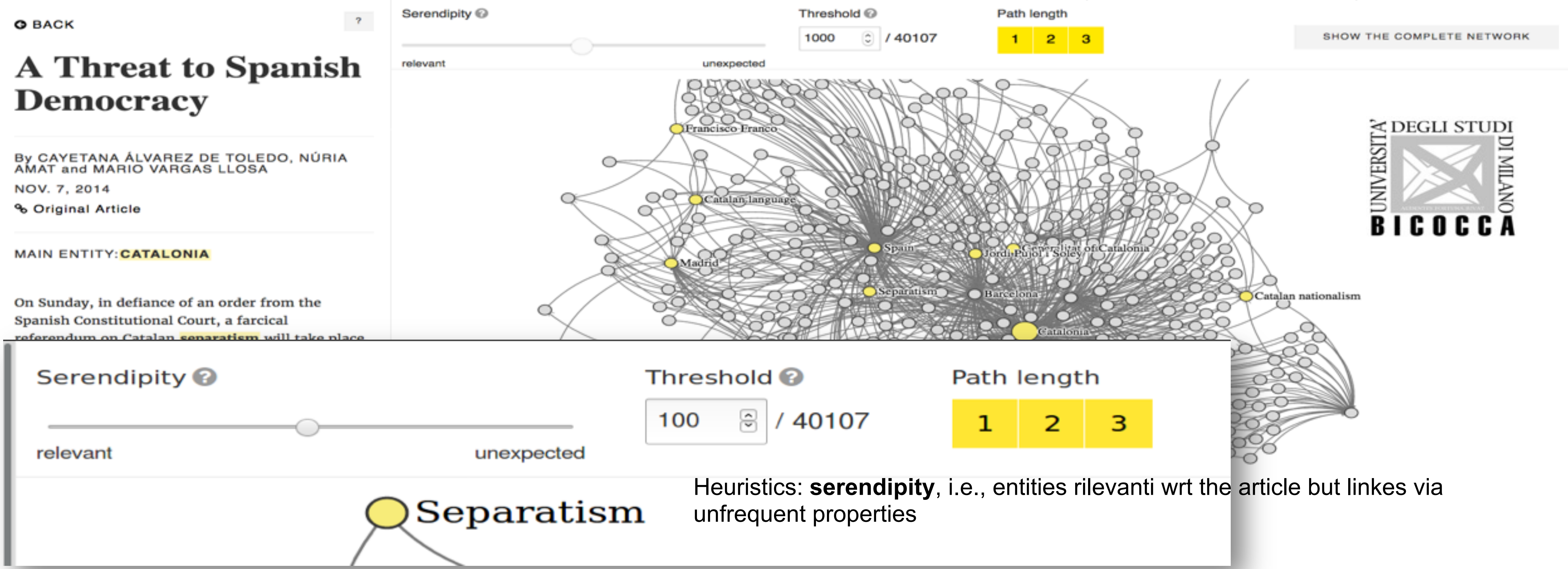


Paths (i.e., *semantic associations*) extracted from the DBpedia KG because deemed to be *relevant* for the articles

Catalonia	leaderName >	Artur Mas i Gavarró	< president	Democratic Convergence of Catalonia	ideology >	Catalan nationalism
Catalonia	< isPartOf	Esplugues de Llobregat	leaderParty >	Socialists' Party of Catalonia	ideology >	Catalan nationalism
Catalonia	< isPartOf	Barcelona	< headquarter	Republican Left of Catalonia	ideology >	Catalan nationalism
Catalonia	< isPartOf	Vilallonga del Camp	leaderParty >	Republican Left of Catalonia	ideology >	Catalan nationalism
Catalonia	< isPartOf	Sant Feliu Sasserra	leaderParty >	Republican Left of Catalonia	ideology >	Catalan nationalism
Catalonia	< isPartOf	Riudaura	leaderParty >	Republican Left of Catalonia	ideology >	Catalan nationalism
Catalonia	< nationality	Enric Prat de la Riba	party >	Regionalist League	ideology >	Catalan nationalism
Catalonia	< isPartOf	Alcanar	leaderParty >	Republican Left of Catalonia	ideology >	Catalan nationalism

Which Entities and Relations are More Interesting?

More popular…
Less popular ... (more unexpected?)



Heuristics: **serendipity**, i.e., entities rilevanti wrt the article but linkes via unfrequent properties

But different users are interested in different information! (Bianchi&al, 2017)
Clue: learning to rank most relevant information for the individual user ☺

Abstraction and Visualization of Large Graphs

A screenshot analyzing a dataset with
about 600 millions triples
Summary view (A)

Provides **high-level contextual information**, such as:

- data size,
- current selection size, and, optionally,
- statistics (i.e., data peeking)

By C. Batini (Keynote @ICWE 2017)

P. C. Wong *et al.*, "A visual analytics paradigm enabling trillion-edge graph exploration," *2015 IEEE 5th Symposium on Large Data Analysis and Visualization (LDAV)*, Chicago, IL, 2015, pp. 57-64.

Schema-based Profiling of Large KGs to Understand their Content

ABSTAT: profiling large KGs (e.g., DBpedia 2015-10)

How many entities of type Company?
Which types are linked to each other?
Which properties are used to describe the entities and with which frequency?

Occurrence of types (as minimal types) and properties

	subject type (occurrences)	predicate (occurrences)	object type (occurrences)	frequency	instances	Max subjs-obj	Avg subjs-obj	Min subjs-obj	Max subj-objs	Avg subj-objs	Min subj-objs
filter	dbo:Company	predicate	object								
	dbo:Company (51898)	DTP foaf:name (5002938)	rdfs:Literal (12948466)	52391	64155	200	1	1	16	1	1
	dbo:Company (51898)	OP foaf:homepage (362016)	owl:Thing	42861	52699	27	1	1	11	1	1
	dbo:Company (51898)	OP dbo:industry (50100)	owl:Thing	42408	50093	1750	10	1	15	1	1
	dbo:Company (51898)	DTP dbo:foundingYear (87412)	xmls:gYear (3469462)	42131	49085	1159	99	1	8	1	1
	dbo:Company (51898)	OP dbo:predecessor (78232)	dbo:Person (611330)	8	11	1	1	1	1	1	1

Minimal Patterns: there exist entities that have Company as *minimal type, which* are linked to literals that have gYear as *minimal type* by the property foundingYear

Cardinality descriptors: max/avg/min number of different subjects associated with a same object (and vice versa)

Frequency and instances: how many times this pattern occurs as minimal type pattern and as a non minimal type pattern (considers frequency of more specific pattterns)

Check ABSTAT online: http://abstat.disco.unimib.it/ - recommendation: use it for exercises with SPARQL queries

Profiling & Learning to Explore Knowledge – SMARTGraphX Project Proposal

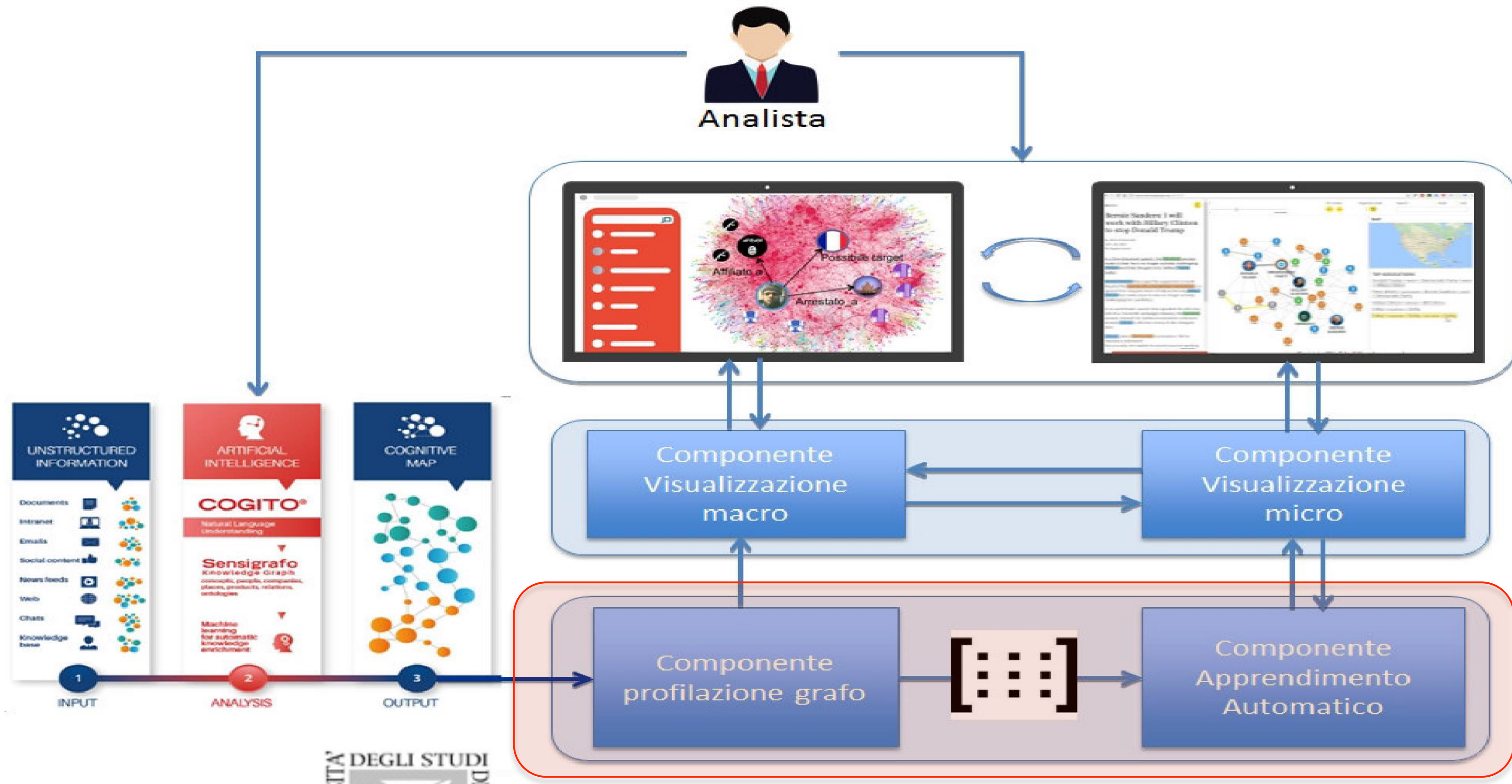
Profiling and exploration of large KGs

- Ontology-base profiling: understand the content of the KG
- Entity-specific profiling and heuristics to define relevant elements

Learning what is more relevant for the user

Idea: combining all the above concepts to support information filtering and recommendation

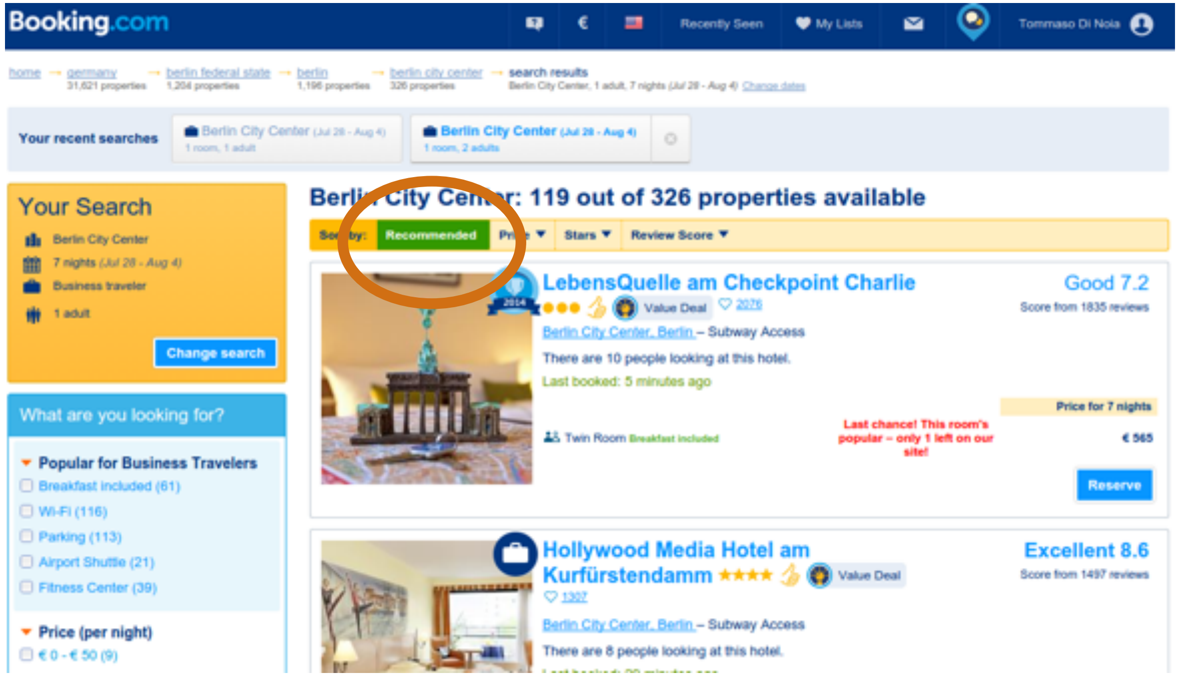
Visualization: important but not sufficient ingredient

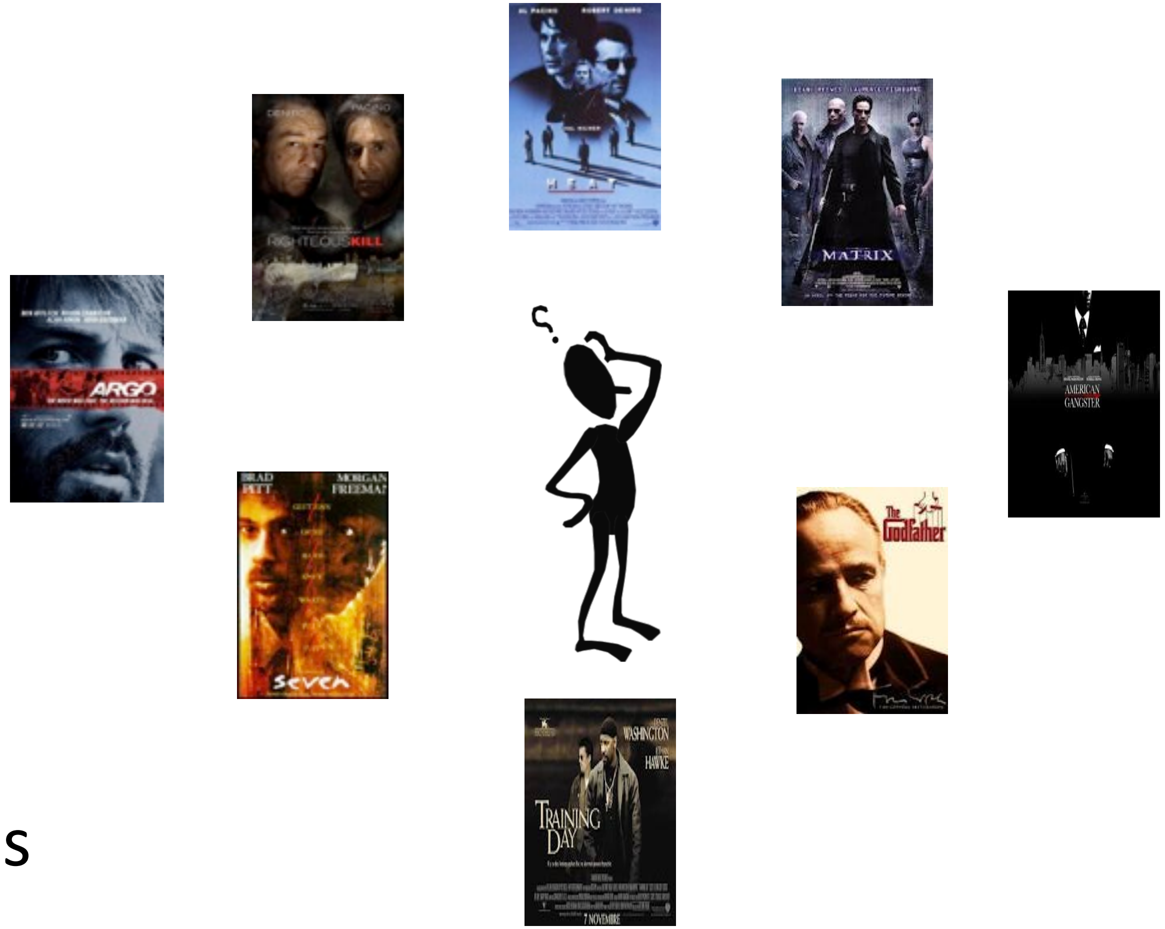


UNIVERSITÀ DEGLI STUDI DI MILANO BICOCCA

Recommender Systems and the Role of Content

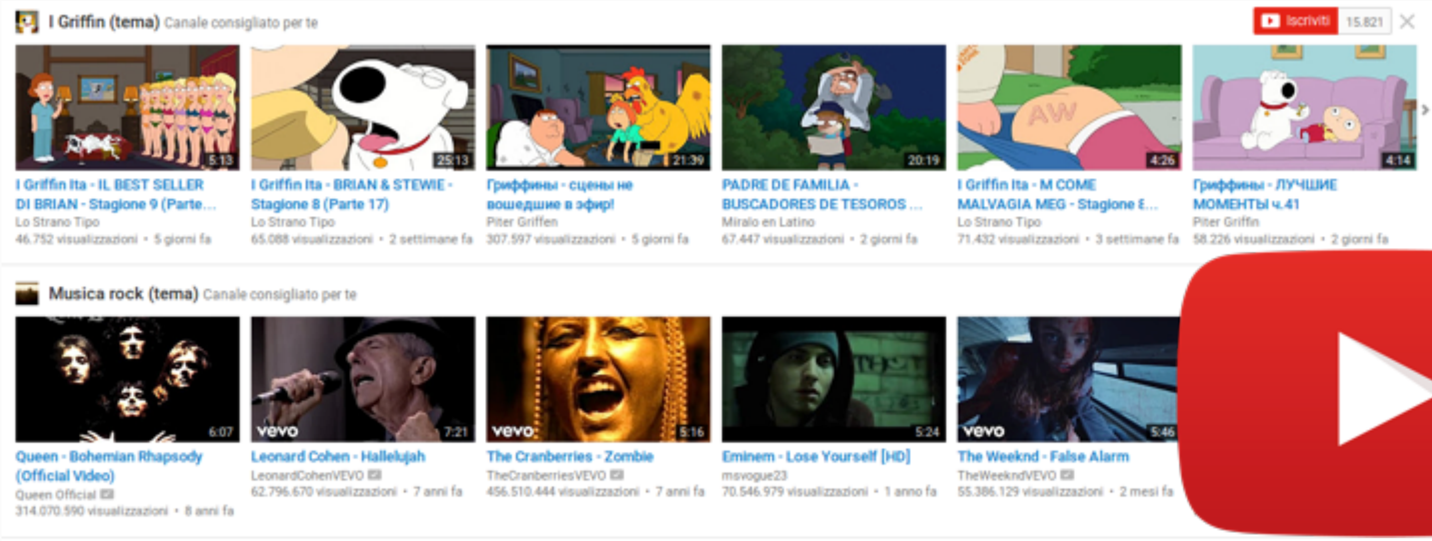
Recommender Systems





- Help users in dealing with information/choice overload
- Help to match users with items

Examples of Recommender Systems



COMPUTERS Vs HUMANS

Select music that people can enjoy

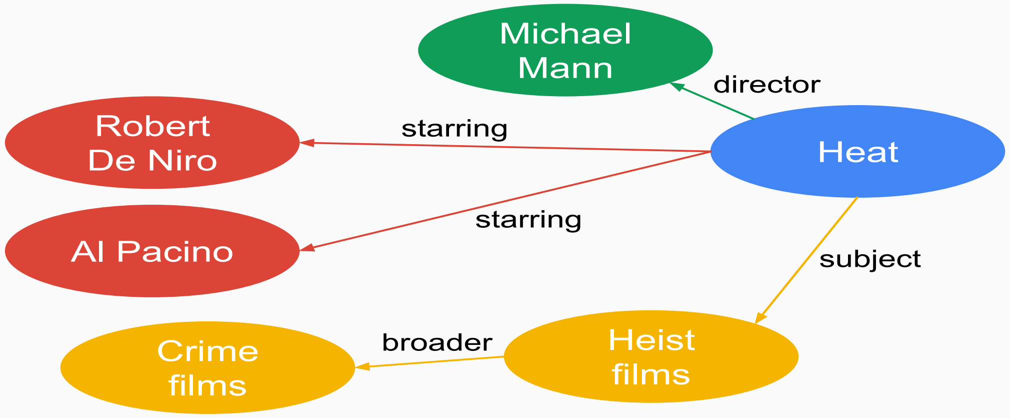
Oldest DJ in the world (age 75)

Maximum of 7MIL songs at 95
by superficially listening music 12h/day

DJ Algoriddim (age 1)

Deep analysis of 40MIL songs NOW

Recommender Systems & Semantics



	STARRING		DIRECTOR	SUBJECT+BROADER	
HEAT	Robert De Niro	Al Pacino	Micheal Mann	Heist films	Crime films

Recommend to a user *u* items that are similar to items that have been liked by users similar to *u*

Knowledge Bases may be used to compute similarity between items
(see presentation about similarity)

Similarity

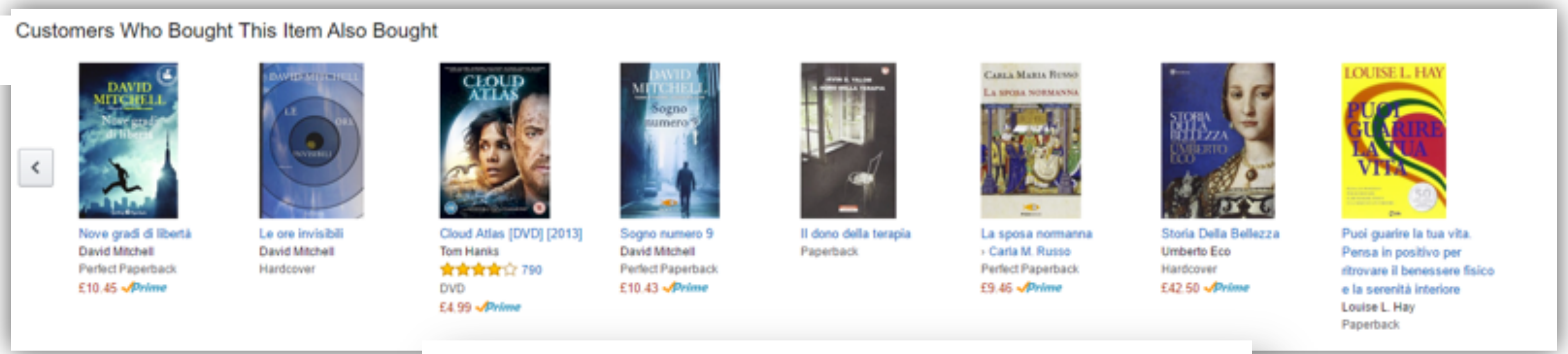
Key point: **evaluate the similarity** between users and/or items

Two approaches:

Collaborative-filtering based: recommend items to the users searching for other users with similar profile (user-based o item-based)

Content-based: recommend items to a user based on the description of the item and the interests of the user (similarity between the recommended item and items that the user has selected in the past)

Why do we need content?



Several Recommender Systems **perfectly work** **without using any content!** (e.g.Amazon)

Recommending New Movies: Even a Few Ratings Are More Valuable Than Metadata

István Pilászy *
Dept. of Measurement and Information Systems
Budapest University of Technology and Economics
Magyar Tudósok krt. 2.
Budapest, Hungary
pila@mit.bme.hu

Domonkos Tikk *,†
Dept. of Telecom. and Media Informatics
Budapest University of Technology and Economics
Magyar Tudósok krt. 2.
Budapest, Hungary
tikk@tmit.bme.hu

ABSTRACT

The Netflix Prize (NP) competition gave much attention to collaborative filtering (CF) approaches. Matrix factorization (MF) based CF approaches assign low dimensional feature vectors to users and items. We link CF and content-based filtering (CBF) by finding a linear transformation that transforms user or item descriptions so that they are as close as possible to the feature vectors generated by MF for CF.

We propose methods for explicit feedback that are able to handle 140 000 features when feature vectors are very sparse. With movie metadata collected for the NP movies we show that the prediction performance of the methods is comparable to that of CF, and can be used to predict user preferences on new movies.

We also investigate the value of movie metadata compared to movie ratings in regards of predictive power. We compare

1. INTRODUCTION

The goal of recommender systems is to give personalized recommendation on items to users. Typically the recommendation is based on the former and current activity of the users, and metadata about users and items, if available.

There are two basic strategies that can be applied when generating recommendations. Collaborative filtering (CF) methods are based only on the activity of users, while content-based filtering (CBF) methods use only metadata. In this paper we propose hybrid methods, which try to benefit from both information sources.

The two most important families of CF methods are matrix factorization (MF) and neighbor-based approaches. Usually, the goal of MF is to find a low dimensional representation for both users and movies, i.e. each user and movie is associated with a feature vector. Movie metadata (which

(ACM RecSys 2009, by Neflix Challenge winners)

Collaborative Filtering and **Matrix Factorization** are state of the art techniques for implementing Recommender Systems

Content **can tackle some issues** of **collaborative filtering**

Collaborative filtering - User based

	The Matrix	Titanic	I love shopping	Argo	Love Actually	The hangover
Tommaso	5	1	2	4	3	?
Enrico	2	4	5	3	5	2

How much similar Tommaso and Enrico are?

We need to consider only the movies that both users have evaluated and evaluate how much correlated their scores are

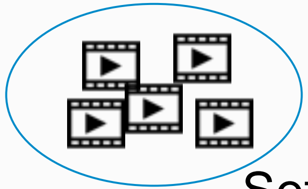
$$sim(T,E) = \frac{(5-3)\cdot(2-3.8)+(1-3)\cdot(4-3.8)+\cdots+(3-3)\cdot(5-3.8)}{\sqrt{(5-3)^2+(1-3)^2+\cdots+(3-3)^2}\cdot\sqrt{(2-3.8)^2+(4-3.8)^2+\cdots+(5-3.8)^2}}$$

$$sim(T,E) \approx -0.73$$

If sim(T,E) > 0, T and E are *directly correlated* (*positively correlated)*
If sim(T,E) = 0, T and E are *not correlated*
If sim(T,E) < 0, T and E are *inversely correlated* (*negatively correlated)*

Similarity ≈ correlation

Collaborative filtering - User based



Set of all movies

	The Matrix	Titanic	I love shopping	Argo	Love Actually	The hangover
Tommaso	5	1	2	4	3	??
Enrico	2	4	5	3	5	2
Sean	4	3	2	4	1	3
Natasha	3	5	1	5	2	4
Valentina	4	4	5	3	5	2

We look at the rows of the score matrix

Correlation
Pearson's corr:
values in [-1,1]

$$sim(u_i, u_j) = \frac{\sum_{x \in X}(r_{u_i,x} - \overline{r_{u_i}}) \cdot (r_{u_j,x} - \overline{r_{u_j}})}{\sqrt{\sum_{x \in X}(r_{u_i,x} - \overline{r_{u_i}})^2} \cdot \sqrt{\sum_{x \in X}(r_{u_j,x} - \overline{r_{u_j}})^2}}$$

Prediction:

$$\tilde{r}(u_i, x') = \overline{r_{u_i}} + \frac{\sum_{u_j} sim(u_i, u_j) \cdot (r_{u_j,x'} - \overline{r_{u_j}})}{\sum_{u_j} sim(u_i, u_j)}$$

Another coefficient:
Jaccard Index

$$sim(u_i, u_j) = \frac{|R_{u_i}(X) \cap R_{u_j}(X)|}{|R_{u_i}(X) \cup R_{u_j}(X)|}$$

Collaborative filtering - Item based

	The Matrix	Titanic
Tommaso	5	1
Enrico	2	4
Sean	4	3
Natasha	3	5
Valentina	4	4

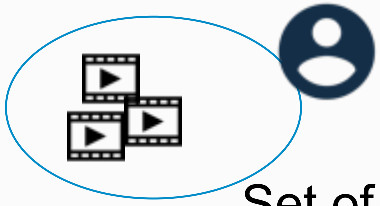
Tommaso has watched both The Matrix and Titanic. How much similar these movies are?

$$sim(T,E) = \frac{(5\cdot 1)+(2\cdot 4)+(4\cdot 3)+(3\cdot 5)+(4\cdot 4)}{\sqrt{5^2+2^2+4^2+3^2+4^2}\cdot\sqrt{1^2+4^2+3^2+5^2+4^2}}$$

$$sim(T,E) \approx 0.82$$

Tommaso gave opposite scores to the two movies. However, the similarity between the movies is high, if we consider scores given by all the users

Collaborative filtering - Item based



Set of all movies

	The Matrix	Titanic	I love shopping	Argo	Love Actually	The hangover
Tommaso	5	1	2	4	3	??
Enrico	2	4	5	3	5	2
Sean	4	3	2	4	1	3
Natasha	3	5	1	5	2	4
Valentina	4	4	5	3	5	2

We look at the columns of the score matrix

Cosine Similarity:

$$sim(\vec{x_i}, \vec{x_j}) = \frac{x_i \cdot x_j}{|\vec{x_i}| \cdot |\vec{x_j}|} = \frac{\sum_u r_{u,x_i} \cdot r_{u,x_j}}{\sqrt{\sum_u r_{u,x_i}^2} \cdot \sqrt{\sum_u r_{u,x_j}^2}}$$

Prediction:

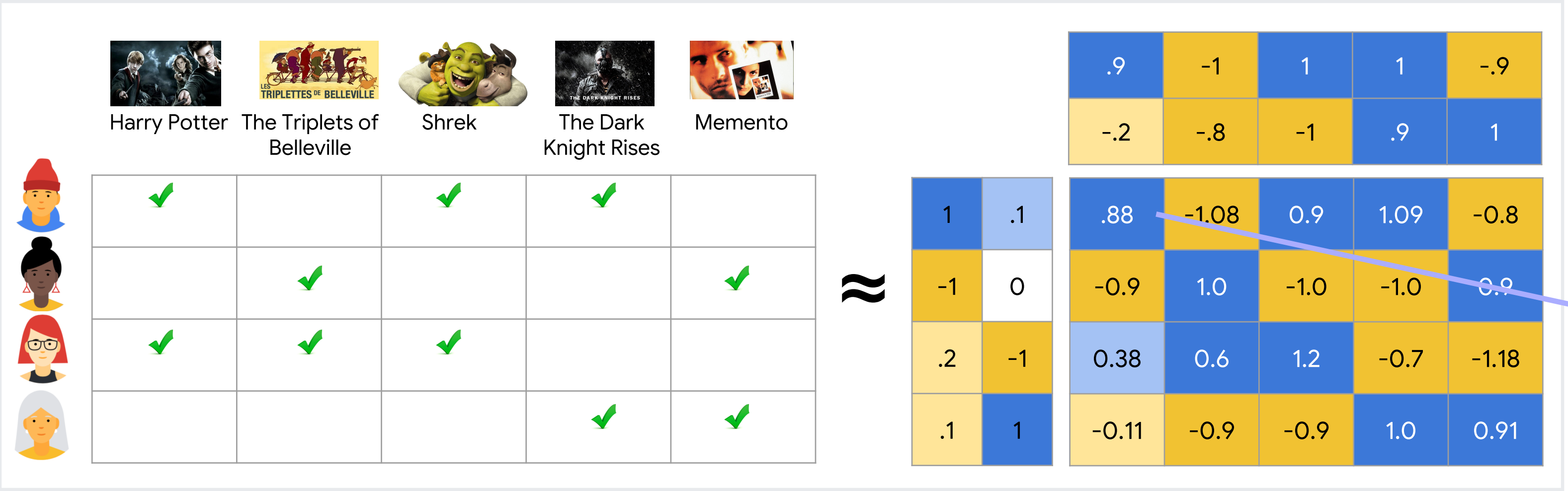
$$\tilde{r}(u_i, x') = \frac{\sum_{x \in X_{u_i}} sim(\vec{x}, \vec{x}') \cdot r_{x,u_i}}{\sum_{x \in X_{u_i}} sim(\vec{x}, \vec{x}')}$$

Matrix Factorization for Collaborative Filtering

Matrix factorization is a simple embedding model. Given the feedback matrix A $\in R^{m \times n}$, where m is the number of users (or queries) and n is the number of items, the model learns:

- A user embedding matrix $U \in \mathbb{R}^{m \times d}$, where row i is the embedding for user i.
- An item embedding matrix $V \in \mathbb{R}^{n \times d}$, where row j is the embedding for item j.

Learn item and user vectors (dense representations of fixed dimension *d*) that approximate R^{m*n}



$[1, 0.1][0.9, -0, 2]$

$(1 * 0.9) + (0.1 * -0.2) = 0.9 - 0.02$

The embeddings are learned such that the product UV^T is a good approximation of the feedback matrix A. Observe that the (i, j) entry of $U.V^T$ is simply the dot product $\langle U_i, V_j \rangle$ of the embeddings of user i and item j, which you want to be close to $A_{i,j}$.

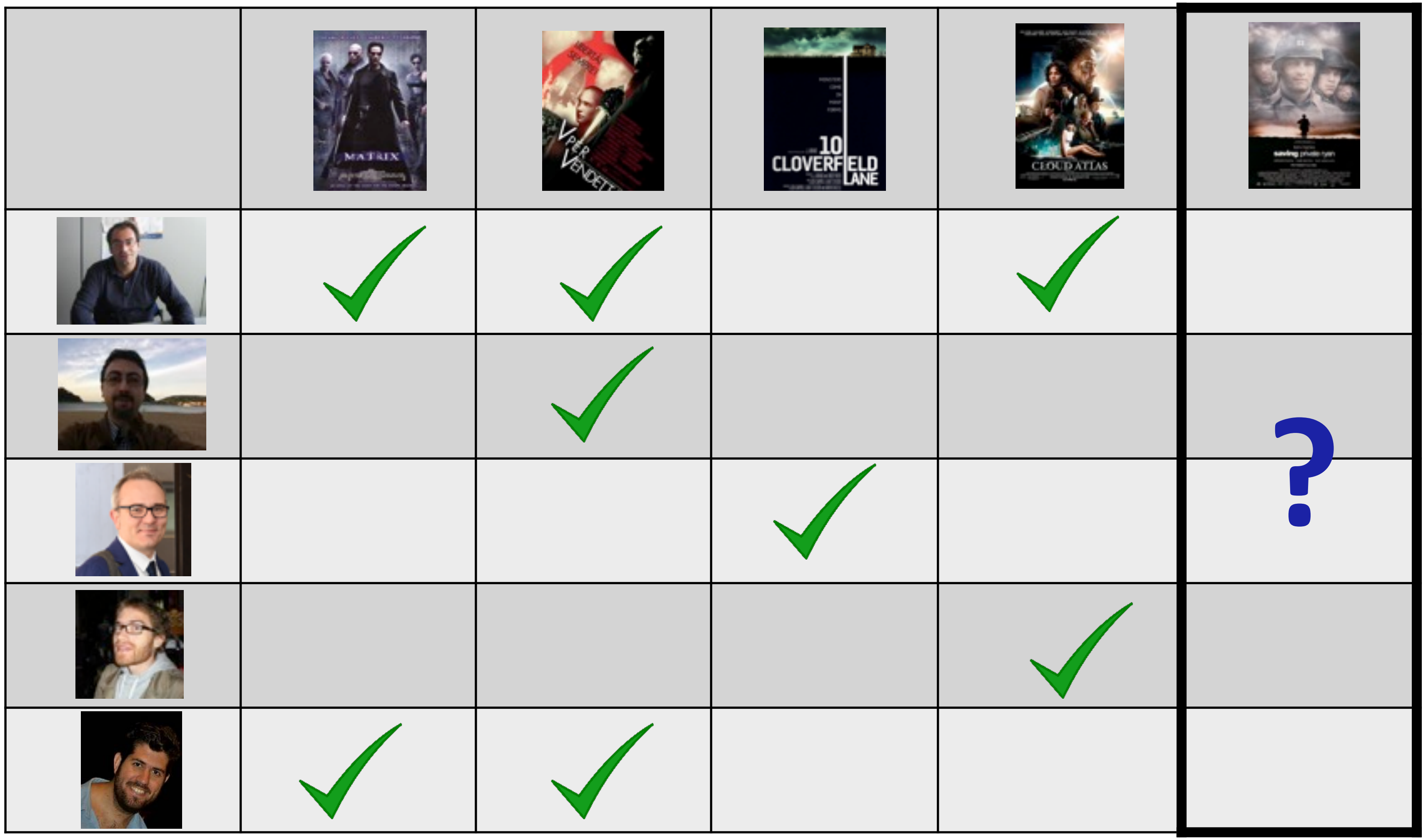
From: https://developers.google.com/machine-learning/recommendation/collaborative/matrix

Why do we need content?

Collaborative Filtering issues: sparsity

29

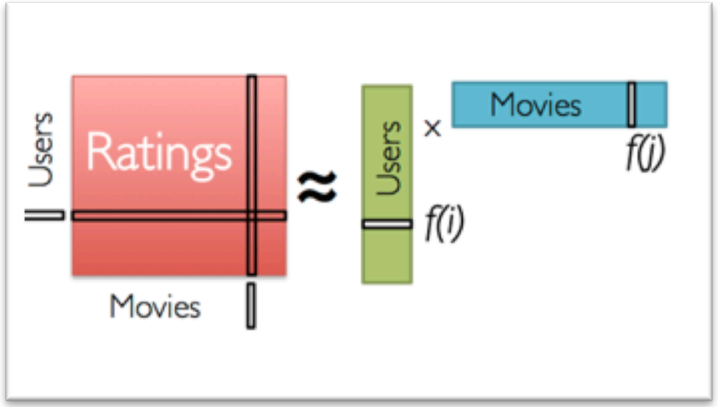
Why do we need content?



Collaborative Filtering issues: <u>new item problem</u>

30

Why do we need content?



Who knows the «customers who bought…»?

Collaborative Filtering issues: poor explanations!

31

Collaborative filtering - Recap

We use opinions of other users that have common interests

Problems:

- sparsity of the score matrix (low probability of finding users who have evaluated the same item)
- new item / user
- gray sheep: a user with a profile different from the profiles of every other user makes it difficult to find item to recommend to him (no information on similar users)
- opaque recommendations

Linked Open Data for Semantics-Aware Recommender Systems

Pierpaolo Basile, Cataldo Musto — SWAP researchgroup, Università degli Studi di Bari Aldo Moro

Tommaso Di Noia, Paolo Tomeo — SisInf Lab, Politecnico di Bari

tutorial@ESWC 2017 – Portoroz (Slovenia) – 29/05/2017

https://semanticrecsys.wordpress.com/

Agenda

Why?

Why do we need **intelligent information access**?
Why do we need **content**?
Why do we need **semantics**?

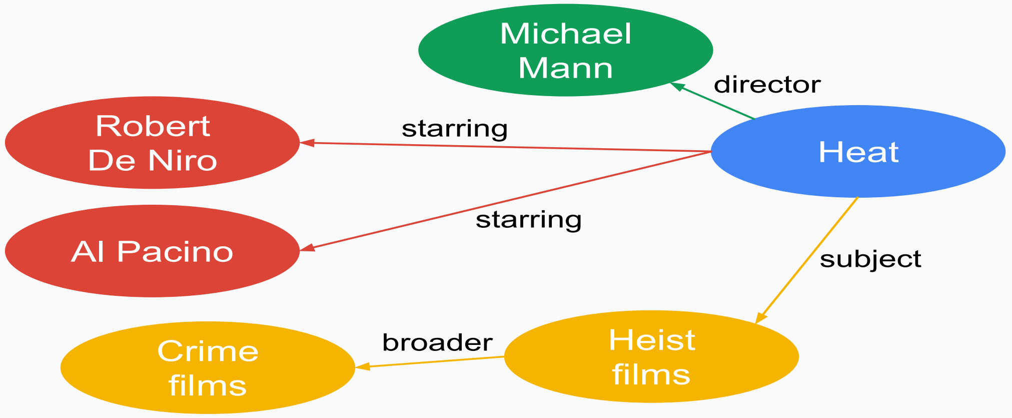
How?

How to **introduce semantics**?
Basics of **Natural Language Processing**
Encoding **exogenous semantics** (top-down approaches)
Encoding **endogenous semantics** (bottom-up approaches)

What?

Recommender Systems based on **Distributional Semantics**
Recommender Systems on **Entity Linking techniques**
Recommender Systems based on **Linked Open Data**
Explaining Recommendations through **Linked Open Data**

Semantic Recommender Systems



	STARRING		DIRECTOR	SUBJECT+BROADER	
HEAT	Robert De Niro	Al Pacino	Micheal Mann	Heist films	Crime films

Recommend to a user *u* items that are similar to items that have been liked by users similar to *u*

KGs may be used to compute similarity between items
(see presentation about similarity)

Content based

Attention on item descriptions (descriptive attributes – features)

Different methods to compute the recommendation:

- from Information Retrieval (e.g., cosine similarity)
- from machine learning to learn users' interests from interaction history

Advantages: we can recommend new items, even if not evaluated by any user

Content based - Limitations

Difficult to provide good recommendations if no discriminant information is available

Often based on keywords or features to describe objects

What if

- the user selects only items that use few keywords?
- keywords not sufficient to discriminate?
- features are insufficient? (open problem: how to select features?)

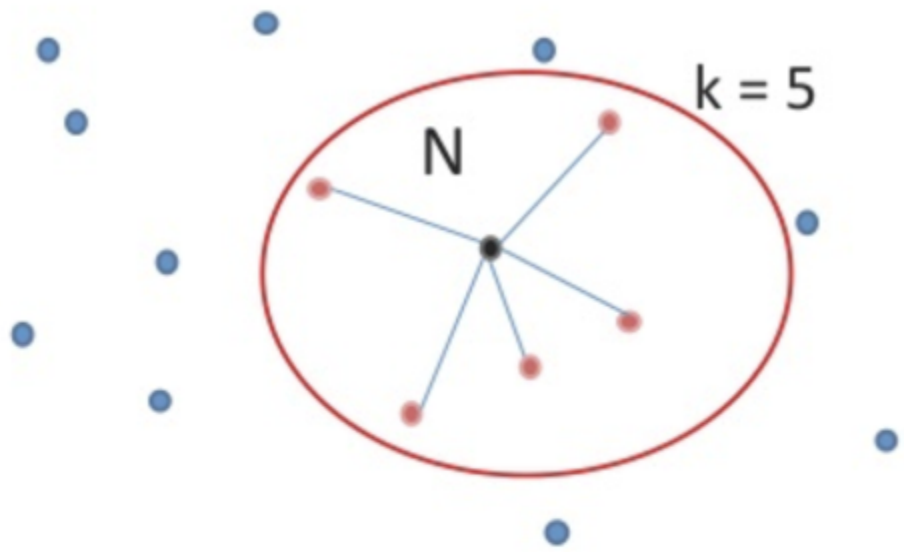
Content-based Semantic Recommendations

- Basic item KNN recommender system
- Given an user *u* a non rated item *i*, the rating of *i* is predicted by:

$$r^*(u,i) = \frac{\sum_{j \in N(i) \cap r(u)} sim(i,j) \cdot r(u,j)}{\sum_{j \in N(i) \cap r(u)} sim(i,j)}$$

where:

- $N(i)$ = neighbors of the non rated item i
- $r(u)$ = the items rated by the user u,
- $r(u,j)$ = the rating value given to the item i by the user u

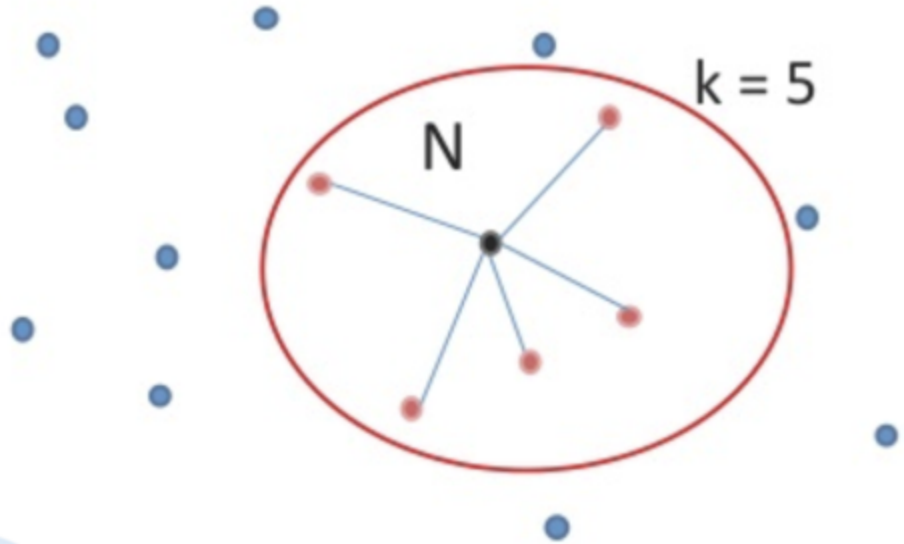


Similarity functions:

- Jaccard
- Graph kernels
- Cosine similarity in a vector space
- … several variants

… all based on subgraphs built using certain properties

Content-based Semantic Recommendations



Rich item descriptions

	STARRING		DIRECTOR	SUBJECT+BROADER	
Heat	Robert DeNiro	Al Pacino	Michael Mann	Heist films	Crime films

Similarity functions:

- Jaccard
- Cosine similarity in a vector space
- Graph kernels
- … several variants

… all based on subgraphs built using certain properties

Jaccard Similarity

- Jaccard similarity
  - Measures the values of the selected features that are shared between two items

$$jaccard(i,j) = \frac{|N(i) \bigcap N(j)|}{|N(i) \bigcup N(j)|}$$

Which properties should be considered? → Feature Selection Problem

Vector-based Similarity with Weighted Properties

Vector-based model … (for each property) cosine similarity between the property weight vectors of each item

STARRING	Al Pacino (v1)	Robert De Niro (v2)	Brian Dennehy (v3)
Righteous Kill (m1)	✓	✓	✓
Heat (m2)	✓	✓	✗

Righteous Kill (m1)	$w_{v1,x1}$	$w_{v2,x1}$	$w_{v3,x1}$
Heat (m2)	$w_{v1,x2}$	$w_{v2,x2}$	0

$$sim_{starring}(\vec{x_i},\vec{x_j}) = \frac{w_{v_1,x_i} * w_{v_1,x_j} + w_{v_2,x_i} * w_{v_2,x_j} + w_{v_3,x_i} * w_{v_3,x_j}}{\sqrt{w^2_{v_1,x_i} + w^2_{v_2,x_i} + w^2_{v_3,x_i}} * \sqrt{w^2_{v_1,x_j} + w^2_{v_2,x_j} + w^2_{v_3,x_j}}}$$

$$\alpha_{starring} * sim_{starring}(\vec{x_i},\vec{x_j}) +$$

$$\alpha_{director} * sim_{director}(\vec{x_i},\vec{x_j}) +$$

$$\alpha_{subject} * sim_{subject}(\vec{x_i},\vec{x_j}) +$$

$$\ldots =$$

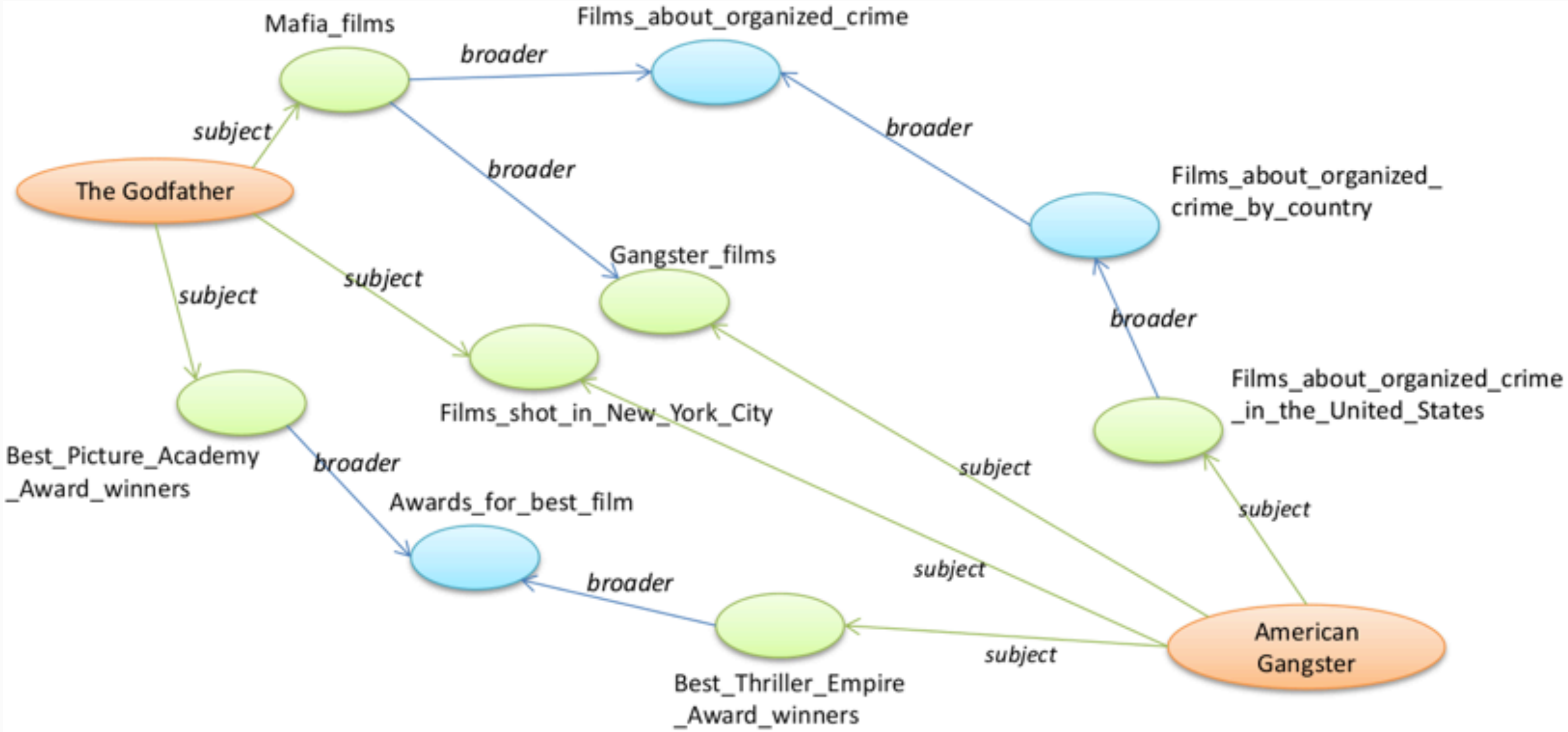
$$sim(\vec{x_i},\vec{x_j})$$

$$w_{AlPacino,Heat} = tf_{AlPacino,Heat} \cdot idf_{AlPacino} \longleftarrow tf \in \{0,1\}$$

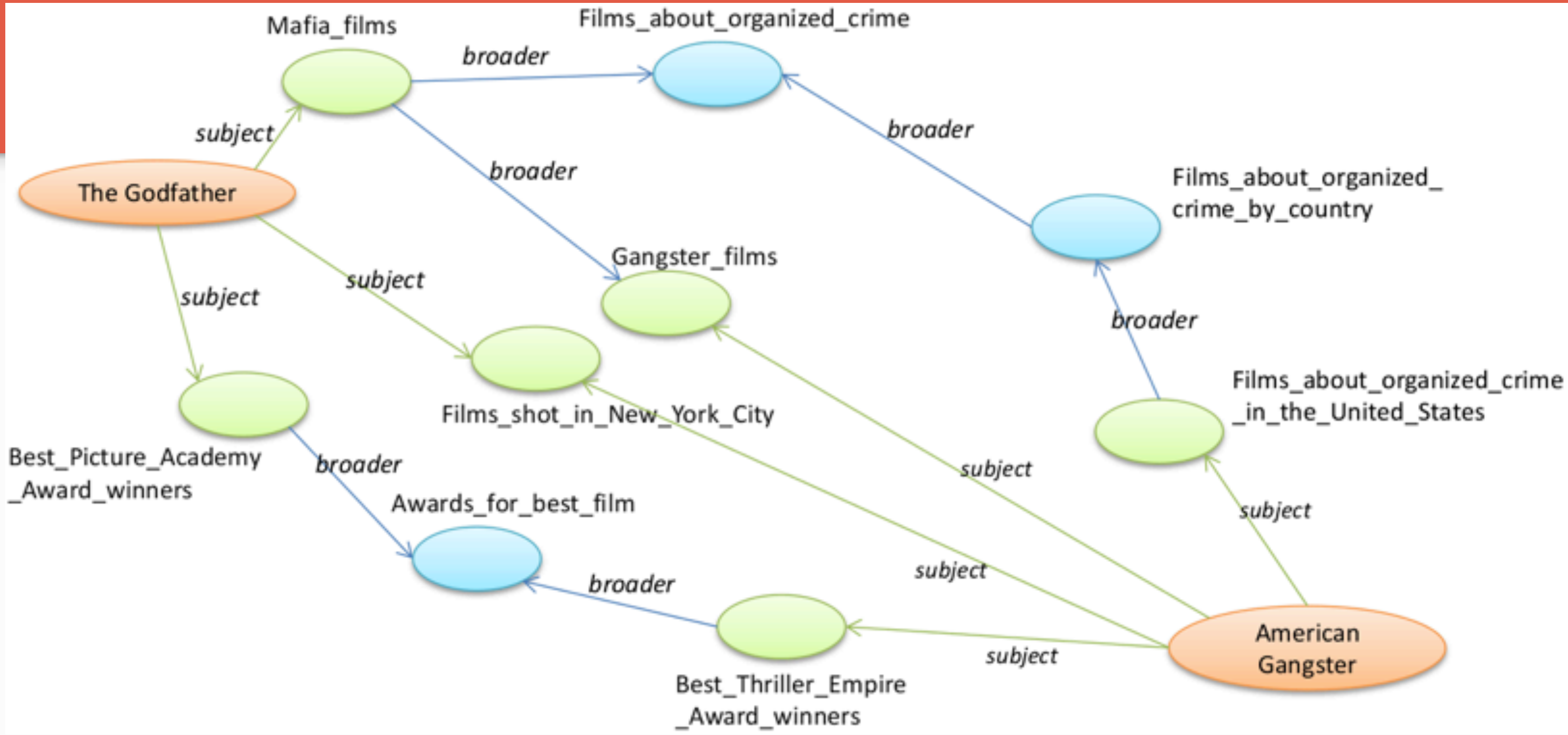
Neiborhood-based Representations

Graph-based representation of the entities (h-hop Item Neighborhood Graph)

Each entity is identify by itself and by its neighbors



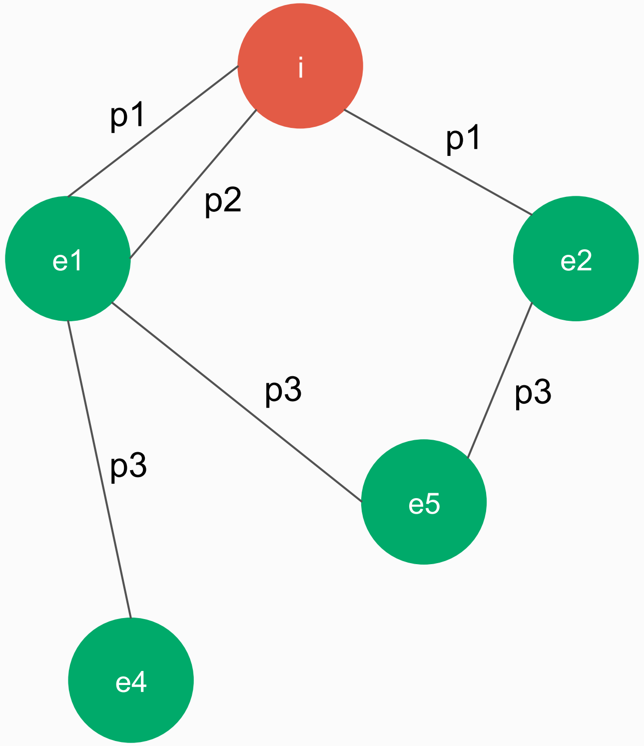
h-Hop Item Neighborhood Graph



For entity i, its neighborood graph is

- $G^h(i) = (E^h(i), P^h(i))$
- $E^h(i)$ = set of entities reachable in at most h hops from i (shortest path)
- $\hat{E}^h(i)$ = set of entities reachable in exactly h hops from i
- $P^h(i)$ = set of edges reachable in at most h hops from i (shortest path)

h-Hop Item Neighborhood Graph



Generation of sets *E* and *P* for *h* = 1 and *h* = 2

$$E^1(i) = \{e_1, e_2\} \qquad P^1(i) = \{p_1(i,e_1), p_2(i,e_1), p_1(i,e_2)\}$$
$$E^2(i) = \{e_1, e_2, e_4, e_5\} \qquad P^2(i) = \{p_1(i,e_1), p_2(i,e_1), p_1(i,e_2), p_3(e_1,e_4), p_3(e_1,e_5), p_3(e_2,e_5)\}$$
$$\widehat{E}^1(i) = \{e_1, e_2\} \qquad \widehat{P}^1(i) = \{p_1(i,e_1), p_2(i,e_1), p_1(i,e_2)\}$$
$$\widehat{E}^2(i) = \{e_4, e_5\} \qquad \widehat{P}^2(i) = \{p_3(e_1,e_4), p_3(e_1,e_5), p_3(e_2,e_5)\}$$

h-Hop Neighborhood-based Graph Kernel

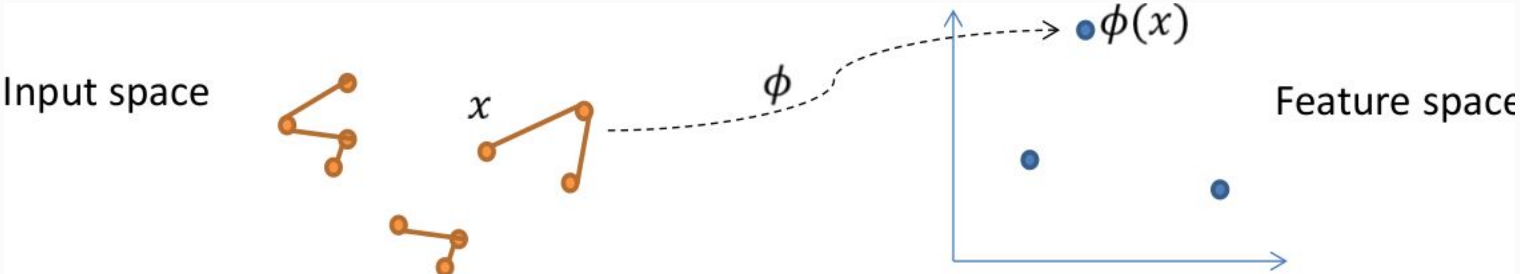
- Towards graph-based similarity

$$k_{G^h}(i,j) = \langle \phi_{G^h}(i), \phi_{G^h}(j) \rangle$$

  - This vector product essentially counts entities common in two graphs, weighted by their importance
- We need to map the input in a feature space (so as to support – after normalization – vector-based similarity)

- Each w_{i,e_j} represents the importance of the entity e_j in the neighborhood of the *i*th entity

$$\phi_{G^h}(i) = (w_{i,e_1}, w_{i,e_2}, \ldots w_{i,e_m}, \ldots, w_{i,e_t})$$



Feature Weighting for Similarity Between h-Hop Neiborhood Graphs

- Compute feature e_m for entity *i*:

$$w_{i,e_m} = \sum_{l=1}^{h} \alpha_l \cdot c_{\widehat{P}^l(i),e_m}$$

- where:

$$c_{\widehat{P}^l(i),e_m} = |\{p_k(e_n, e_m) \mid p_k(e_n, e_m) \in \widehat{P}^l(i) \wedge e_m \in \widehat{E}^l(i)\}|$$

- i.e., number of edges that involve the node e_m in the *l* hop, i.e., occurrences of e_m in the neighborhood of distance *l*
- *α is a* parameter for distance-based weighting: e.g., favor importance of entities closers to *i*

Computing the Similarity

- Different entity graphs may have different dimensions. If we normalize them, we can obtain cosine similarity [1]
- Otherwise
  - Possible application of algorithms for graph similairty
    - (e.g. Maximal Common Subgraph - MCS [2])

[1] Sergio Oramas, Vito Claudio Ostuni, Tommaso Di Noia, Xavier Serra, and Eugenio Di Sciascio. 2016. Sound and Music Recommendation with Knowledge Graphs. ACM Trans. Intell. Syst. Technol. 8, 2, Article 21 (October 2016), 21 pages.
[2] Oramas, Sergio, et al. "A semantic-based approach for artist similarity." *Proceedings of the 16th International Society for Music Information Retrieval (ISMIR) Conference;* International Society for Music Information Retrieval (ISMIR), 2015.

New Approaches for Item-to-item Similarity

RDF2Vec: RDF graph embeddings and their applications

Article type: Research Article

Authors: Ristoski, Petar[a; *] | Rosati, Jessica[b; c] | Di Noia, Tommaso[b] | De Leone, Renato[c] | Paulheim, Heiko[a]

Affiliations: [a] Data and Web Science Group, University of Mannheim, B6, 26, 68159, Mannheim. E-mails: petar.ristoski@informatik.uni-mannheim.de, heiko@informatik.uni-mannheim.de | [b] Polytechnic University of Bari, Via Orabona, 4, 70125, Bari. E-mails: jessica.rosati@poliba.it, tommaso.dinoia@poliba.it | [c] University of Camerino, Piazza Cavour 19/f, 62032, Camerino. E-mails: jessica.rosati@unicam.it, renato.deleone@unicam.it

Correspondence: [*] Corresponding author. E-mail: petar.ristoski@informatik.uni-mannheim.de.

Abstract: Linked Open Data has been recognized as a valuable source for background information in many data mining and information retrieval tasks. However, most of the existing tools require features in propositional form, i.e., a vector of nominal or numerical features associated with an instance, while Linked Open Data sources are graphs by nature. In this paper, we present RDF2Vec, an approach that uses language modeling approaches for unsupervised feature extraction from sequences of words, and adapts them to RDF graphs. We generate sequences by leveraging local information from graph sub-structures, harvested by Weisfeiler–Lehman Subtree RDF Graph Kernels and graph walks, and learn latent numerical representations of entities in RDF graphs. We evaluate our approach on three different tasks: (i) standard machine learning tasks, (ii) entity and document modeling, and (iii) content-based recommender systems. The evaluation shows that the proposed entity embeddings outperform existing techniques, and that pre-computed feature vector representations of general knowledge graphs such as DBpedia and Wikidata can be easily reused for different tasks.

Keywords: Graph embeddings, linked open data, data mining, document semantic similarity, entity relatedness, recommender systems

Petar Ristoski, Jessica Rosati, Tommaso Di Noia, Renato De Leone, Heiko Paulheim: **RDF2Vec: RDF graph embeddings and their applications.** Semantic Web 10(4): 721-752 (2019)

Word2vec (explained in next section) on random walks on the graph

Evaluate the Recommender System

Information Retrieval measures:

- precision: relevant items that have been retrieved over the total number of retrieved items
- recall: relevant items that have been retrieved over the total number of relevant items

Ranking quality measures: nDCG (for non binary problems)

- Items associated with graded relevance, e.g., [1,2,3] (remark → we can define an ideal ranking)
- Discounted Cumulative Gain (at position *p*): comulate gain (gain = rel_i = graded relevance of a retrieved item) and discount this gain as long as the rank position decreases
- normalize DCG by comparing it with Ideal DCG, i.e., DCG of the ideal ranking

$$\mathrm{DCG_p} = \sum_{i=1}^{p} \frac{rel_i}{\log_2(i+1)}$$

$$\mathrm{nDCG_p} = \frac{DCG_p}{IDCG_p}$$

Measures from Machine Learning :

- Mean Absolute Error (MAE), Root Mean Square Error (RMSE)

Evaluation: Several Quality Dimensions

Relevance: recommend items that are relevant to the user

Novelty: recommend items that are not known to the user. Often a trade-off between novelty and accuracy

Serendipity: recommend items that are not expected (novelty + surprise). Risk of recommending non relevant items but allows to widen user's interests

Diversity: recommend items that are different increase the likelihood to recoomend at least one good item (+ experience)

Evaluation: Several Quality Dimensions

Relevance: recommend items that are relevant to the user

Novelty: recommend items that are not known to the user. Often a trade-off between novelty and accuracy

Serendipity: recommend items that are not expected (novelty + surprise). Risk of recommending non relevant items but allows to widen user's interests

Diversity: recommend items that are different increase the likelihood to recoomend at least one good item (+ experience)

Explainability: recommend items that can be explained to (and understood by) the user

KGs for Explainable Recommender Systems

International Semantic Web Conference
ISWC 2019: The Semantic Web – ISWC 2019 pp 38-56 | Cite as

How to Make Latent Factors Interpretable by Feeding Factorization Machines with Knowledge Graphs

Authors   Authors and affiliations

Vito Walter Anelli, Tommaso Di Noia, Eugenio Di Sciascio, Azzurra Ragone, Joseph Trotta

Conference paper
First Online: 17 October 2019

Part of the Lecture Notes in Computer Science book series (LNCS, volume 11778)

Abstract

Model-based approaches to recommendation can recommend items with a very high level of accuracy. Unfortunately, even when the model embeds content-based information, if we move to a latent space we miss references to the actual semantics of recommended items. Consequently, this makes non-trivial the interpretation of a recommendation process. In this paper, we show how to initialize latent factors in Factorization Machines by using semantic features coming from a knowledge graph in order to train an interpretable model. With our model, semantic features are injected into the learning process to retain the original informativeness of the items available in the dataset. The accuracy and effectiveness of the trained model have been tested using two well-known recommender systems datasets. By relying on the information encoded in the original knowledge graph, we have also evaluated the semantic accuracy and robustness for the knowledge-aware interpretability of the final model.

Vito Walter Anelli, Tommaso Di Noia, Eugenio Di Sciascio, Azzurra Ragone, Joseph Trotta: **How to Make Latent Factors Interpretable by Feeding Factorization Machines with Knowledge Graphs**. ISWC (1) 2019: 38-56

Best Student Paper @ ISWC 2019

KGs for Explainable Recommender Systems

Linked open data-based explanations for transparent recommender systems

Cataldo Musto, Fedelucio Narducci, Pasquale Lops, Marco de Gemmis, Giovanni Semeraro

Show more

https://doi.org/10.1016/j.ijhcs.2018.03.003

Highlights

- We design the main components of an *algorithm-agnostic* framework to generate *natural language* explanations;
- We propose a methodology to extract descriptive (direct and indirect) properties about the items, and we use these properties to feed a graph-based explanation model;
- We define a scoring function to rank these explanation patterns and we use the most relevant ones to generate a template-based natural language explanation;
- We validate our methodology by carrying out a large user study (N = 680) in three different domains, as movies, books and music;
- We integrate our methodology in a conversational *recommender system* implemented as a Telegram Bot.

Cataldo Musto, Fedelucio Narducci, Pasquale Lops, Marco de Gemmis, Giovanni Semeraro: **Linked open data-based explanations for transparent recommender systems**. Int. J. Hum.-Comput. Stud. 121: 93-107 (2019)

Cracking the Feature Selection Problem for KG-based RecSys

European Semantic Web Conference
ESWC 2018: The Semantic Web pp 128-144 | Cite as

Using Ontology-Based Data Summarization to Develop Semantics-Aware Recommender Systems

Authors | Authors and affiliations

Tommaso Di Noia, Corrado Magarelli, Andrea Maurino, Matteo Palmonari, Anisa Rula

Conference paper
First Online: 03 June 2018

Properties that optimize content-based similarity
selected with our KG profiling framework ABSTAT

Abstract

In the current information-centric era, recommender systems are gaining momentum as tools able to assist users in daily decision-making tasks. They may exploit users' past behavior combined with side/contextual information to suggest them new items or pieces of knowledge they might be interested in. Within the recommendation process, Linked Data have been already proposed as a valuable source of information to enhance the predictive power of recommender systems not only in terms of accuracy but also of diversity and novelty of results. In this direction, one of the main open issues in using Linked Data to feed a recommendation engine is related to feature selection: how to select only the most relevant subset of the original Linked Data thus avoiding both useless processing of data and the so called "curse of dimensionality" problem. In this paper, we show how ontology-based (linked) data summarization can drive the selection of properties/features useful to a recommender system. In particular, we compare a fully automated feature selection method based on ontology-based data summaries with more classical ones, and we evaluate the performance of these methods in terms of accuracy and aggregate diversity of a recommender system exploiting the top-k selected features. We set up an experimental testbed relying on datasets related to different knowledge domains. Results show the feasibility of a feature selection process driven by ontology-based data summaries for Linked Data-enabled recommender systems.

Di Noia, T., Magarelli, C., Maurino, A., Palmonari, M., & Rula, A. (2018, June). **Using ontology-based data summarization to develop semantics-aware recommender systems**. In European Semantic Web Conference (pp. 128-144). Springer, Cham.

If interested, check the annexed presentation

Recent Approaches & Deep Learning

- Hybrid approaches: collaborative-filtering + content-based
- Embedding relevant features via Deep Learning
- Several side interesting problems:
    - Check the slides of the Music Recommendation Tutorial on the website